

Ulaş Uçrak

+90 542 541 03 65 | ulasucrak@gmail.com | linkedin.com/in/ulasucrak | github.com/ulasucrak

SUMMARY

Computer Engineering graduate from Çankaya University (2026) with a 3.91/4.00 GPA, ranked 1st in the Faculty of Engineering. Software experience at SDT, ASELSAN and Leonardo Türkiye in C++/Qt desktop software, embedded firmware, Python/FastAPI and React/Node.js. Software lead of the team placed 4th worldwide at CanSat 2025. Winner of the Cursor Ankara Hackathon.

EDUCATION

Çankaya University | *B.S. in Computer Engineering* 2022 – 2026
GPA: 3.91/4.00, ranked 1st in the Faculty of Engineering
TED Ankara College | *High School Diploma* 2022

EXPERIENCE

SDT Uzay & Savunma Teknolojileri | *Candidate Engineer (Part-time)* Sep 2025 – Jun 2026

- Developed and maintained C++ software components for Qt-based desktop applications used in defense and space projects.
- Also worked on embedded systems software. Held the role part-time alongside the final year of the degree.

ASELSAN | *Intern, REHİS (Radar & Electronic Warfare Systems)* Aug 2025 – Sep 2025

- Designed and implemented a custom I2C-based communication protocol between STM32 boards in embedded C.
- Contributed to hardware-software integration on STM32 microcontrollers, and tested and debugged embedded software.

Çankaya University | *Undergraduate Teaching Assistant (Part-time)* 2023 – 2025

- Helped students in C programming labs debug their code and understand the material, and graded assignments.

Leonardo Türkiye | *Intern* Jun 2024 – Jul 2024

- Developed the complete firmware for an ESP32-based IoT device: sensor data collection, wireless communication, OTA updates.
- Built a Node.js server interface for HTTP-based firmware updates, using JSON parsing to compare firmware versions.
- Optimized asynchronous HTTP requests to improve the reliability of device-server communication.

Bilkent Cyberpark | *Web Development Intern* Jan 2024 – Feb 2024

- Built a personal CV website in React during a four-week web development internship.

PROJECTS & COMPETITIONS

CanBEE: CanSat Model Satellite | *Software Team Lead* 2025

4th worldwide, 1st among foundation universities, CanSat Competition 2025 (American Astronautical Society, NASA-sponsored)

- Led the software team and defined the software architecture, including real-time integration with the ground station.
- Developed the embedded software and designed the ground control interface; coded and tested the core mission functions.
- Supervised integration and tuned the software during a two-week on-site field-test campaign in the United States.

FitStake: Web3 "Stake-to-Fit" Platform | *1st place, Cursor Ankara Hackathon: Blockchain Edition*

- Built a platform in 5 hours where users stake SOL and earn rewards for completing verified workouts.
- Stack: MediaPipe (real-time exercise detection), Rust/Anchor smart contracts on Solana (on-chain rewards), React frontend.

Helma AI (HealthMate): AI-Powered Elderly Care Assistant

- Built the LLM reasoning component of a multimodal elderly-care system and the Python FastAPI service that serves it.
- Designed Turkish conversational prompts and structured JSON outputs for daily-routine and health inference.

Understanding Bitcoin Sentiments: Insights from Tweets and Users

- Used machine learning to analyze the sentiment of Bitcoin-related tweets, its correlation with price trends and user influence.
- Preprocessed datasets, trained predictive models and built a real-time sentiment analysis application.

BEHSAT (TEKNOFEST model satellite): Designed the GUI and implemented embedded code for the university's model satellite team.

ReadLens: C++ accessibility tool that reads text in photographs aloud for visually impaired users (Tesseract OCR, eSpeak).

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, C#, Swift, HTML, CSS
Frameworks & Libraries: Qt, React, Node.js, FastAPI, .NET, Electron, MediaPipe, Tesseract OCR
Databases: MySQL, Firebase, Supabase, ChromaDB
Embedded: STM32, ESP32, embedded C, I2C, OTA firmware updates
Tools & Platforms: Git, Linux (Ubuntu), Windows
AI / ML: pandas, NumPy, scikit-learn, NLP (text preprocessing, sentiment analysis), LLM prompt design, structured JSON outputs
Spoken Languages: Turkish (Native), English (C1), German (A2)

LEADERSHIP

Computer Engineering Society, Çankaya University | *President (Volunteer)* 2024 – 2025

- Organized events and seminars, coordinated with companies and sponsors, and assigned tasks within the board.